

Q> Print :-

```
4
4 3
4 3 2
4 3 2 1
```

```
#include <iostream>
using namespace std;
```

```
int main () {
    int n;
    cin >> n;
```

```
for (int i = 1 ; i <= n ; i++) {
```

```
    for (int j = n ; j >= n - i + 1 ; j--) {
```

```
        cout << j << " ";
```

```
    }
    cout << endl;
```

```
    }
    return 0;
```

Q> Print the following pattern, if $n = 5$

```

      *
     *
    *
   *
  *
 *

```

Analysis →

Star	Row	Space
1	1	4
2	2	3
3	3	2
4	4	1
5	5	0

Logic →

(i) Row = 1 to Row ≤ 5

(ii) Print Space, (5 - row) times or (n - row) times

(iii) Print Star row times

(iv) Increment Row by 1.

Code →

```
#include <iostream>
using namespace std;
```

```
int main() {
```

```
    int n;
```

```
    cin >> n;
```

```
    for (int i=1; i<=n; i++) {
```

```
        for (int j=1; j<=(n-i); j++) { // For Printing Spaces
```

```
            cout << "  "; // Double Space given
```

```
        }
        for (int k=1; k<=i; k++) { // For printing stars
```

```
            cout << "* "; // An additional Space given
```

```
        }
        cout << endl;
```

```
    }
    return 0;
```

Q> Print Pattern if $n=4$

```

      1
     2 2
    3 3 3
   4 4 4 4
```

Logic →

(i) Row=1 to Row ≤ 4

(ii) Print Space $(n-\text{row})$ times

(iii) Print row no., row times

(iv) Increment row by 1

Analysis →

Space	Row	Print
$4-1=3$	1	1
$4-2=2$	2	2
$4-3=1$	3	3
$4-4=0$	4	4

Space is printed $(n-\text{row})$ times

Row no gets Printed, row times

Code →

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
```

```
for (int i=1; i<=n; i++) {
```

```
    for (int j=1; j<=(n-i); j++) { // Prints Spaces
```

```
        cout << "  "; // Double Space
```

```
    }
```

```
    for (int k=1; k<=i; k++) { // Prints Numbers
```

```
        cout << "X ";
```

```
        cout << i << " ";
```

```
    }
```

```
    cout << endl;
```

```
}
```

```
return 0;
```

```
}
```

Q) If $n=5$, Print Pattern:

```

      A B
     A B C
    A B C D
   A B C D E
  A B C D E
```

Logic →

(i) Row = 1 to Row ≤ 5

(ii) Print Space (n-row) times

(iii) Print ('A') to ('A' + row - 1)

(iv) Increment row by one.

Code →


```
#include <iostream>
using namespace std;
```

```
int main () {
    int n;
    cin >> n;

    for (int i=1; i<=n; i++) {
        for (int j=1; j<=n-i; j++) { // Prints Space
            cout << " "; // Double Space
        }
        for (int k=65; k<=i+64; k++) { // Prints Alphabet
            cout << (char)k << " ";
        }
        cout << endl;
    }
    return 0;
}
```

Second Approach :-

```
for (char name='A'; name<='A'+row-1; name++) {
    cout << name << " ";
}
```

Q> If $n=5$, Print the pattern:-

```

    1
  2 1
 3 2 1
4 3 2 1
5 4 3 2 1
```

Logic $\Rightarrow$

(i) Row = 1 to Row ≤ 5

- (ii) Print Space (n-row) Times
- (iii) Print Row to 1
- (iv) Increment Row by 1.

Code →

```
#include <iostream>
using namespace std;
```

```
int main () {
```

```
    int n;
```

```
    cin >> n;
```

```
    for(int i=1 ; i<=n; i++) {
```

```
        for(int j=1 ; j<=(n-i); j++) {
```

// Prints Space

```
            cout << " ";
```

Double

```
        }
        for(int k=i; k>=1; k--) {
```

// Prints numbers

```
            cout << k << " ";
```

```
        }
        cout << endl;
```

```
    }
    return 0;
```

```
}
```

Q> If n = 5 Print the Pattern: -

```

      1
     1 2
    1 2 3
   1 2 3 4
  1 2 3 4 5

```

Logic →

- (i) Row 1 to Row ≤ 5
- (ii) Print Space (n-row) Times
- (iii) Print 1 to Row
- (iv) Increment Row by 1

Code →

```
#include <iostream>
using namespace std;
```

```
int main () {
```

```
    int n;
```

```
    cin >> n;
```

```
    for(int i=1 ; i<=n; i++) {
```

```
        for(int j=1 ; j<=(n-i); j++) { // Prints Space
```

```
            cout << "  "; // Double Space
```

```
        }
```

```
        for(int k=1 ; k<=i ; k++) { // Prints numbers
```

```
            cout << k << " ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

Q> If $n=5$, Print pattern:-

```

      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * *

```

Analysis →

Row	Space	Star
1	4	1
2	3	3
3	2	5
4	1	7
5	0	9